

BROOKINGS, OR, USA

WMO: 725985

Lat: 42.074N

Lon: 124.290W

Elev: 459

StdP: 14.45

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24267

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	33.7	35.8	17.4	13.4	51.7	22.6	17.3	50.4	24.5	53.1	20.6	51.2	2.1	50	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	15.5	85.9	63.2	80.5	60.9	75.0	59.6	65.5	80.5	63.7	74.9	62.2	71.4	8.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	83.0	66.4	60.2	79.1	65.1	58.8	75.1	63.7	30.4	80.5	29.2	75.2	28.1	71.5	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.3	13.4	DB	28.9	95.9	2.3	5.7	27.2	100.0	25.9	103.3	24.6	106.5	22.9	110.7	(4)
(5)			WB	27.2	69.0	2.5	2.4	25.4	70.8	24.0	72.2	22.6	73.6	20.7	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	48.0	47.9	48.0	50.8	54.3	58.5	59.8	59.8	60.2	56.1	51.0	47.5	(6)	
	DBStd	7.27	5.16	5.44	4.70	5.38	5.31	5.88	5.36	4.42	6.38	5.34	4.92	5.01	(7)	
	HDD50	500	99	93	95	47	9	0	0	0	0	6	44	107	(8)	
	HDD65	4298	527	478	528	429	337	218	183	174	176	285	419	544	(9)	
	CDD50	1781	36	35	32	71	143	254	302	304	306	194	76	28	(10)	
	CDD65	104	0	0	0	3	6	22	21	13	31	8	0	0	(11)	
	CDH74	1365	0	1	1	36	98	292	324	165	375	71	2	0	(12)	
	CDH80	518	0	0	0	5	31	97	160	56	155	14	0	0	(13)	
(14) Wind	WSAvg	4.6	4.6	5.3	5.5	5.1	5.1	5.0	3.5	3.3	4.0	4.3	4.8	5.1	(14)	
(15) Precipitation	PrecAvg	77.0	12.8	10.0	9.1	7.1	3.9	1.9	0.4	0.7	1.2	4.8	11.0	14.6	(15)	
	PrecMax	124.4	26.3	21.3	17.3	16.8	9.8	6.1	1.7	3.2	5.4	9.0	26.3	31.0	(16)	
	PrecMin	53.2	3.9	3.3	4.2	2.4	0.0	0.1	0.0	0.0	0.0	0.3	3.4	4.3	(17)	
	PrecStd	18.3	5.9	5.2	3.8	3.8	2.3	1.5	0.4	0.8	1.3	2.6	5.5	6.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.9	70.0	67.9	79.4	85.4	89.1	95.0	87.9	93.3	81.3	70.4	64.5	(19)	
		MCWB	52.6	54.9	53.1	56.5	60.1	62.1	65.2	65.6	63.5	58.1	54.8	49.9	(20)	
	2%	DB	63.2	63.5	62.1	69.8	75.4	82.0	82.4	78.3	83.7	74.6	64.3	60.6	(21)	
		MCWB	50.1	51.2	52.3	55.0	58.2	62.0	63.4	62.3	62.7	56.9	53.1	50.3	(22)	
	5%	DB	57.5	58.6	58.6	63.3	69.2	75.6	73.4	72.1	77.2	69.9	60.9	57.0	(23)	
		MCWB	50.7	49.9	51.3	52.5	57.2	60.2	61.5	61.5	60.6	55.6	53.6	51.2	(24)	
	10%	DB	55.1	55.2	55.3	59.1	63.7	70.0	68.1	67.4	70.9	65.1	57.6	54.8	(25)	
		MCWB	51.0	50.3	50.2	51.4	55.2	58.4	60.3	60.6	59.4	54.7	53.2	50.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	57.1	57.4	59.3	63.8	66.4	69.1	67.1	68.2	63.8	60.3	57.1	(27)	
		MCDB	60.6	62.3	62.3	72.8	77.8	83.3	85.6	82.4	86.0	72.0	64.4	58.5	(28)	
	2%	WB	54.5	54.6	54.8	56.4	61.0	63.7	64.6	64.2	64.8	61.1	57.5	54.8	(29)	
		MCDB	56.9	58.3	58.9	65.8	71.6	77.6	79.3	73.9	78.6	67.9	60.2	56.4	(30)	
	5%	WB	53.4	52.8	53.0	54.3	58.2	61.4	62.6	62.4	62.8	58.9	55.9	53.5	(31)	
		MCDB	55.7	56.1	56.3	60.7	67.0	73.4	72.8	69.9	72.1	65.1	58.4	54.9	(32)	
	10%	WB	51.7	51.3	51.5	52.8	55.9	59.3	60.5	60.9	60.9	57.2	54.4	52.0	(33)	
		MCDB	54.5	54.1	54.2	57.6	62.9	68.6	68.0	67.0	68.1	61.5	56.7	53.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	11.6	12.3	13.5	14.0	15.4	14.5	13.4	15.5	13.9	11.5	11.5	(35)	
		MCDBR	17.7	17.4	18.2	21.7	24.8	27.3	26.3	23.7	27.8	23.0	16.9	14.7	(36)	
		MCWBR	10.6	9.7	10.8	11.8	12.4	12.8	12.2	11.6	11.0	10.9	9.4	9.0	(37)	
	5% WB	MCDBR	13.1	12.5	13.8	17.8	21.0	23.5	22.5	19.7	20.9	16.7	12.1	9.4	(38)	
		MCWBR	8.8	8.0	9.1	10.4	11.4	11.9	11.2	10.4	10.5	9.9	8.0	7.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.314	0.314	0.320	0.343	0.348	0.339	0.340	0.339	0.331	0.323	0.310	0.314	(40)	
	taud		2.576	2.584	2.589	2.482	2.511	2.544	2.548	2.566	2.559	2.599	2.612	2.578	(41)	
	Ebn at Noon		267	286	296	294	294	295	294	292	288	278	267	257	(42)	
	Edh at Noon		22	25	28	33	33	32	32	30	28	24	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		541	797	1166	1628	2046	2266	2315	1964	1575	1026	609	443	(44)	
	RadStd		82	111	124	144	122	137	102	112	91	95	72	81	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+4

Nomenclature: See separate page